Multiplicity-Corrected Inverse Probability Scaling for Reward-Proportional Sampling in Large DAGs

Abstract

Many scientific generation problems require sampling diverse, high-reward outcomes rather than finding a single maximizer. Count-based inverse probability scaling (IPS) estimates terminal probability from repetitions within a sampled group. When the terminal space is large relative to the group, repetitions become rare and the correction becomes nearly constant, reducing the update toward GRPO. Replacing terminal probability with trajectory probability does not solve this problem in a directed acyclic graph: terminals reachable by many trajectories receive excess weight. We introduce MIPS-GRPO, which uses the trajectory weight $R(x)q(\tau \mid x)/P_F(\tau)$, where q is a normalized reverse allocation over paths to x. This allocation makes the total expected weight assigned to each terminal equal to $R(x)$, independent of path multiplicity. In a controlled hypergrid, count IPS recovers all modes when repetitions are common but collapses when they are sparse; MIPS-GRPO recovers all modes in both regimes. In the matched five-taxon experiment, GRPO covers 1 of 105 topologies, IPS-GRPO covers 3, PhyloGFN covers 15, and MIPS-GRPO covers all 105. In molecular synthesis, MIPS-GRPO produces $93.59 \pm 2.43\%$ unique valid samples, compared with $74.54 \pm 3.99\%$ for RGFN and less than 0.2% for GRPO and IPS-GRPO. These results indicate that multiplicity correction can reduce mode collapse while preserving reward quality.

1 Introduction

Drug design, program synthesis, and Bayesian inference often have many useful answers. For such problems, returning only the highest-reward terminal is less useful than representing the full reward-weighted set. The desired target is

$$p^*(x) = \frac{R(x)}{Z}, \qquad Z = \sum_x R(x). \tag{1}$$

Expected-return maximization does not target Eq. (1); it rewards moving probability toward the best terminal and therefore encourages outcome-level collapse. IPS cancels the terminal-frequency factor in the policy gradient by dividing reward by terminal probability [Sinha et al., 2026].

Count IPS-GRPO estimates terminal probability from repeats inside a group of G samples. If the effective terminal space K is much larger than G, almost every observed terminal appears once and $\hat{p}(x) = 1/G$. Group normalization removes the common factor, leaving essentially the GRPO update.

A sampled trajectory has a known probability $P_F(\tau)$, suggesting the weight $R(x)/P_F(\tau)$. This works in a tree but not a DAG. If terminal x has $m(x)$ construction paths, that weight assigns it total mass $m(x)R(x)$, confounding reward with reachability. This is central in both applications studied here: a phylogenetic tree has many merge orders and a molecule may have many synthesis trajectories.

We evaluate the method in three settings. A controlled hypergrid separates a regime with informative counts from one with sparse counts. Phylogenetic inference tests a combinatorial space in which many merge orders lead to the same tree. Molecular synthesis tests a different construction graph and a separate flow-based baseline.

Contributions.

- We characterize the repeat limitation of count IPS through the ratio between the effective terminal space and the sampled group size.
- We derive a normalized reverse-allocation correction that removes path multiplicity without requiring repeated terminal outcomes.
- We evaluate the correction in informative-count and sparse-count hypergrid regimes against GRPO, count IPS-GRPO, and GFlowNet TB.
- We study the same effect in phylogenetic inference and molecular synthesis using matched four-method comparisons.

2 Multiplicity-corrected IPS

Let a forward policy generate trajectory $\tau = (s_0, a_1, \ldots, a_T, s_T)$ and terminal $x = s_T$. Its path probability is known,

$$P_F(\tau) = \prod_{t=1}^{T} P_F(a_t \mid s_{t-1}), \tag{2}$$

but the terminal probability is a sum over paths,

$$P_F(x) = \sum_{\tau \in \mathcal{T}(x)} P_F(\tau). \tag{3}$$

Equation (3) becomes intractable in large DAGs.

2.1 Why naive path IPS is wrong

For $w_{\text{naive}}(\tau) = R(x)/P_F(\tau)$, the expected weighted mass assigned to terminal x is

$$\sum_{\tau \in \mathcal{T}(x)} P_F(\tau) w_{\text{naive}}(\tau) = |\mathcal{T}(x)| R(x). \tag{4}$$

The desired reward is multiplied by path count. This is structural bias, not merely high variance.

2.2 Reverse allocation across paths

Let $q(\tau \mid x)$ be any normalized distribution over paths terminating at x: $\sum_{\tau \in \mathcal{T}(x)} q(\tau \mid x) = 1$. Define

$$w(\tau) = \frac{R(x) q(\tau \mid x)}{P_F(\tau)}. \tag{5}$$

Then

$$\sum_{\tau \in \mathcal{T}(x)} P_F(\tau) w(\tau) = R(x) \sum_{\tau \in \mathcal{T}(x)} q(\tau \mid x) = R(x). \tag{6}$$

Table 1: Compared learning signals. Multiplicity correction means that the terminal target does not acquire an extra factor equal to its number of paths.

Method	Learning signal	Needs repeats?
GRPO	group-normalized $R(x)$	no
Count IPS-GRPO	$R(x)/\hat{p}(x)$, group counts	yes
GFlowNet TB	trajectory-balance residual	no
MIPS-GRPO	$R(x)q(\tau \mid x)/P_F(\tau)$, PPO	no

The normalization of q is enough for Eq. (6); a learned reverse model is not required. A uniform backward policy is valid. If $q(\tau \mid x) = P_F(\tau \mid x)$, the weight becomes the ideal $R(x)/P_F(x)$ and is constant across paths to the same terminal. Learning a reverse model may reduce variance, but that is an empirical benefit rather than a correctness requirement.

2.3 Policy update and relationship to GFlowNets

For each on-policy batch we sample trajectories, score them with a frozen reverse policy, form

$$\log w_i = \log R(x_i) + \log q(\tau_i \mid x_i) - \log P_F(\tau_i), \tag{7}$$

normalize weights into advantages, and take a path-level PPO update. When $q = q_\phi$ is fitted, maximum likelihood on observed reverse edges is performed only after scoring the batch. This order avoids using the same batch to increase its own weight.

GFlowNets also target reward-proportional terminal distributions [Bengio et al., 2021, 2023]. Trajectory balance uses a backward policy and learned partition function in a squared balance residual [Malkin et al., 2022]; MIPS-GRPO uses the backward allocation inside an importance weight and updates the forward policy with PPO.

3 Controlled evaluation: informative and sparse counts

3.1 Hypergrid and two regimes

We use a two-dimensional hypergrid with H^2 terminals and four symmetric high-reward modes. Starting at $(0, 0)$, each action increments one coordinate or terminates. The reward landscape follows Sinha et al. [2026], with $R_0 = 0.1$, $R_1 = 0.5$, and $R_2 = 2.0$. Because $p(x) \propto R(x)$ is enumerable, we measure exact empirical ℓ_1 distance to target, modes recovered, and terminal coverage.

At $H = 8, G = 256$, there are 64 terminals and repeats are plentiful ($K/G = 0.25$). At $H = 64, G = 32$, there are 4096 terminals and repeats are rare ($K/G = 128$). The latter is also a multiplicity stress test: terminal (a, b) has $\binom{a+b}{a}$ paths, and the four equal-reward modes differ in path count by roughly 24 orders of magnitude.

3.2 Results

When counts are informative, count IPS-GRPO lowers ℓ_1 from 1.768 for GRPO to 0.226 ± 0.007 and recovers all modes. Count IPS works here because outcomes repeat within each group. MIPS-GRPO also recovers all modes and reaches 0.057 ± 0.007.

When counts are sparse, GRPO and count IPS-GRPO each recover one mode, with $\ell_1 = 1.966$ and 1.844. MIPS-GRPO visits 4056 of 4096 terminals, recovers all modes, and reaches $\ell_1 = 0.262$.

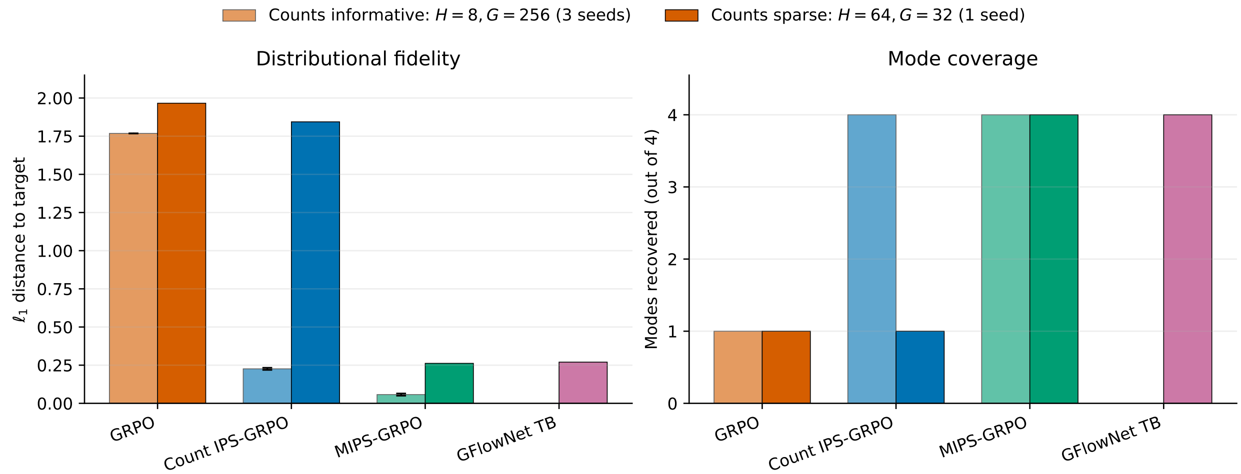


Figure 1: **The count estimator succeeds only while outcomes repeat.** Left: exact ℓ_1 distance to target. Right: modes recovered. Small-grid bars show mean and one standard deviation over three seeds; the large-grid suite is one seed. GFlowNet TB was run only in the large-grid suite.

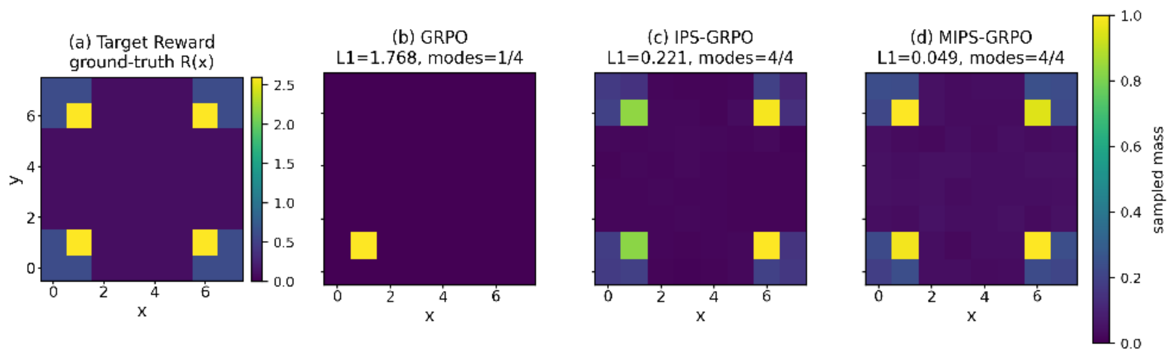


Figure 2: **Positive control: counting works when outcomes repeat.** Terminal distributions for $H = 8, G = 256$ (seed 0). GRPO collapses, count IPS recovers all four modes, and MIPS-GRPO most closely matches the target.

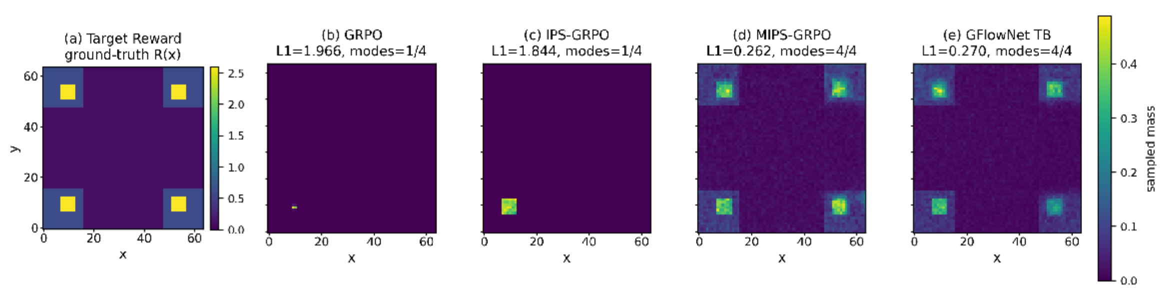
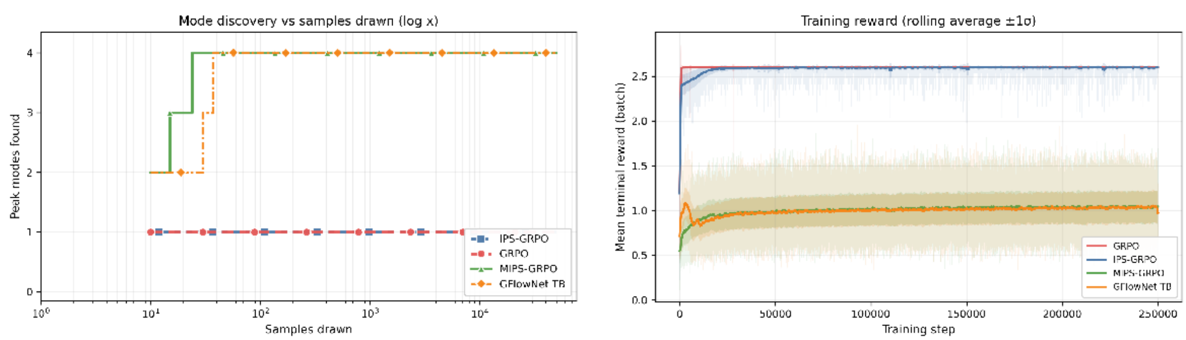


Figure 3: **Stress test: counting fails without repeats.** Terminal distributions for $H = 64, G = 32$ (seed 0). GRPO and count IPS-GRPO recover one mode; MIPS-GRPO and GFlowNet TB recover all four.

Table 2: Hypergrid results in the informative-count and sparse-count regimes. Modes is out of four and ℓ_1 is the unnormalized distance to $p(x) \propto R(x)$, so values are comparable within but not across grid sizes. Small-grid entries are mean $\pm$ standard deviation over three seeds; large-grid entries are seed 0 only.

Method	Modes	ℓ_1	Terminals visited
Counting works: $H = 8, G = 256, K/G = 0.25$			
GRPO	1.0 ± 0.0	1.768 ± 0.000	1/64
Count IPS-GRPO	4.0 ± 0.0	0.226 ± 0.007	64/64
MIPS-GRPO	4.0 ± 0.0	$\mathbf{0.057 \pm 0.007}$	64/64
Counting fails: $H = 64, G = 32, K/G = 128$			
GRPO	1	1.966	17/4096
Count IPS-GRPO	1	1.844	42/4096
MIPS-GRPO	4	**0.262**	4056/4096
GFlowNet TB	4	0.270	4043/4096

GFlowNet TB also recovers all modes and reaches 0.270. MIPS-GRPO and TB have nearly the same error in this seed.



(a) modes found against samples drawn

(b) mean terminal reward during training

Figure 4: **More samples do not fix the collapsed policies.** Left: GRPO and count IPS do not find the missing modes after 50,000 samples. Right: they still achieve high mean reward, so mean reward alone does not measure distribution matching.

GRPO and count IPS still miss modes after 50,000 samples. MIPS-GRPO and TB find all four modes within roughly 10^2 samples (Figure 4). The collapsed policies can still have high mean reward, so reward alone is not enough to measure distribution fit.

4 Application I: Bayesian phylogenetic inference

A phylogenetic tree is built by repeatedly merging subtrees. The same tree can be built in many merge orders. A terminal signature contains its topology and branch lengths. We compare against PhyloGFN, a task-specific trajectory-balance baseline [Zhou et al., 2024].

We first use the five-taxon DS1 reduced benchmark, where all 105 unrooted binary topologies can be enumerated. The matched no-replay evaluation compares GRPO, within-group count IPS-GRPO, MIPS-GRPO, and PhyloGFN using one million final-policy trajectories per method. GRPO collapses to one signature and one topology. IPS-GRPO improves signature support to 66,268 but covers only 3 of 105 topologies. MIPS-GRPO covers all 105 topologies and retains 951,180 distinct signatures; PhyloGFN covers 15 topologies and retains 100,508. Their log–log correlations are similar (0.992 and 0.991), but their topology coverage is not.

Table 3: Matched no-replay five-taxa phylogenetic evaluation, one million self-sampled trajectories per method. All 105 topologies are enumerable. Log–log r is the pathwise probability–reward diagnostic; support metrics are required because a high correlation can coexist with missing topologies. IPS-GRPO denotes the within-group count estimator.

Method	Log–log r	Unique signatures	Topologies
GRPO	—	1	1/105
IPS-GRPO	0.565	66,268	3/105
MIPS-GRPO	**0.992**	**951,180**	**105/105**
PhyloGFN	0.991	100,508	15/105

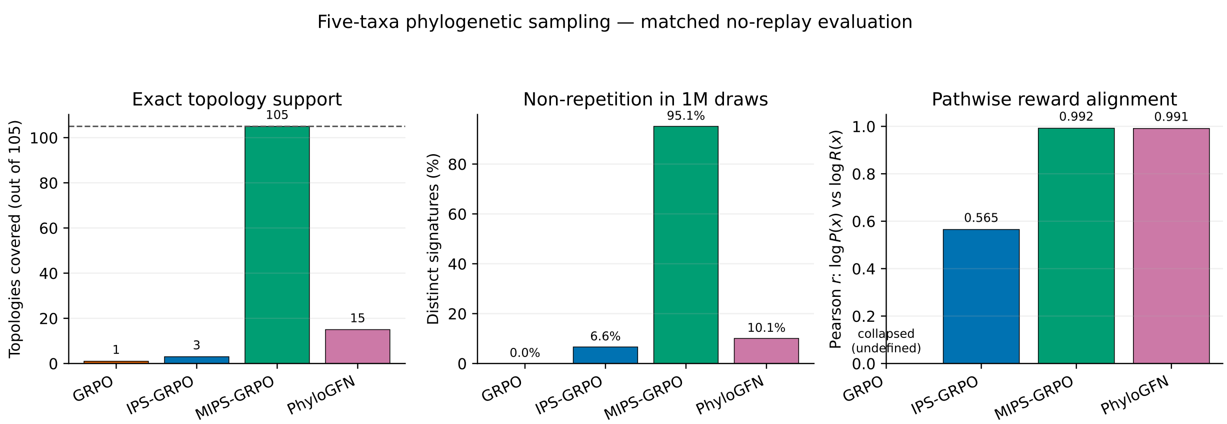


Figure 5: **Matched four-method phylogenetic comparison at five taxa.** GRPO, count-based IPS-GRPO, MIPS-GRPO, and PhyloGFN are evaluated without replay using one million final-policy trajectories each. MIPS-GRPO is the only method to cover all 105 exact topologies and retains 95.1% distinct terminal signatures. PhyloGFN has nearly the same correlation but covers 15 topologies.

5 Application II: full-space molecular synthesis

We test a chemically structured construction DAG using the sEH proxy task. Each method receives 2,500 policy updates (250,000 forward trajectories) and 50,000 independent final-policy samples for each of three seeds. We compare GRPO, count IPS-GRPO, MIPS-GRPO, and RGFN.

MIPS-GRPO produces $93.59 \pm 2.43\%$ unique valid samples, compared with $74.54 \pm 3.99\%$ for RGFN, $0.011 \pm 0.010\%$ for GRPO, and $0.173 \pm 0.286\%$ for count IPS-GRPO. MIPS-GRPO and RGFN have similar mean sampled proxy values (4.281 ± 0.177 and 4.227 ± 0.421) and discover 22.3 ± 17.0 and 14.0 ± 9.2 separated high-proxy modes, respectively. Neither GRPO baseline finds a mode above the threshold in any seed. Mode counts vary across the three seeds, so we do not claim a clear difference between MIPS-GRPO and RGFN on this metric.

Table 4: Full-space sEH proxy task after 2,500 policy updates. Values are mean ± sample standard deviation across three seeds, with 50,000 independent final-policy samples per seed. Unique is the percentage of valid samples that are distinct. A mode has proxy above 7 and maximum Tanimoto similarity at most 0.5 under greedy leader selection.

Method	Valid (%)	Unique (%)	Mean proxy	Modes	Scaffolds > 7
GRPO	100.00 ± 0.00	0.011 ± 0.010	4.261 ± 0.424	0.0 ± 0.0	0.0 ± 0.0
Count IPS-GRPO	100.00 ± 0.00	0.173 ± 0.286	3.816 ± 1.374	0.0 ± 0.0	0.0 ± 0.0
MIPS-GRPO	99.64 ± 0.14	$\mathbf{93.591 \pm 2.427}$	4.281 ± 0.177	$\mathbf{22.3 \pm 17.0}$	32.0 ± 25.5
RGFN	99.94 ± 0.09	74.540 ± 3.994	4.227 ± 0.421	14.0 ± 9.2	32.3 ± 24.2

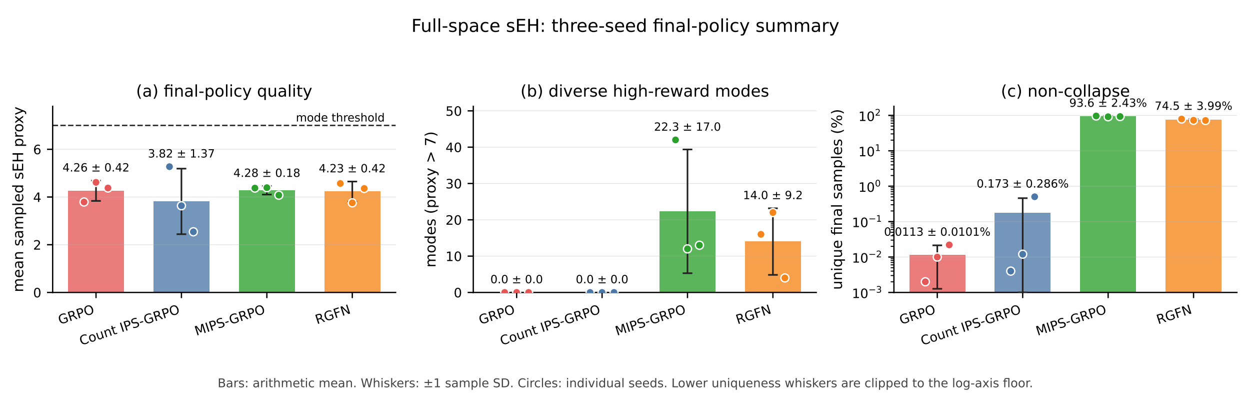


Figure 6: **Full-space sEH final-policy comparison.** Each method was trained for 2,500 updates and evaluated with 50,000 independent samples for each of three seeds. Bars show the arithmetic mean across seeds, whiskers show ±1 sample standard deviation, and circles show individual seeds. Modes have sEH proxy above 7 and maximum Tanimoto similarity at most 0.5 under greedy leader selection. Lower uniqueness whiskers are clipped to the display floor of the logarithmic panel.

6 Summary of findings

Across the controlled and applied experiments, the results follow the expected effect of count sparsity. Count IPS is effective when a sampled group contains repeated outcomes, but its correction becomes nearly constant as repetitions disappear. MIPS-GRPO instead uses path probabilities and a normalized reverse allocation, allowing it to retain broader terminal coverage without relying on counts. The comparisons with GFlowNet TB, PhyloGFN, and RGFN support multiplicity correction as a practical way to reduce collapse, rather than a claim of universal dominance over flow-based methods.

7 Limitations and conclusion

The controlled study evaluates two values of K/G; a denser sweep would characterize the transition more precisely. The phylogenetic correlations are self-sampled diagnostics, so a shared fixed-tree benchmark would provide a stronger cross-method comparison. Equation (5) assumes on-policy data; replay requires an additional correction, and rewards must be positive. The 10-taxon ablation also gives similar effective sample size for fitted and uniform reverse policies, leaving the empirical benefit of learning q_ϕ unresolved.

Within these conditions, MIPS-GRPO provides a count-free correction for reward-proportional sampling when outcomes rarely repeat and multiple paths reach the same terminal. The experiments consistently associate this correction with improved terminal coverage and reduced mode collapse.

A Per-seed hypergrid results

The appendix contains per-seed values, small-scale controls, training curves, and additional diagnostics.

Table A1: Per-seed results for $H = 8, G = 256$ at epoch 1,999, using 10,000 evaluation samples.

Seed	Method	Modes	ℓ_1	Unique terminals
0	GRPO	1	1.768	1
0	Count IPS-GRPO	4	0.221	64
0	MIPS-GRPO	4	0.049	64
1	GRPO	1	1.768	1
1	Count IPS-GRPO	4	0.223	64
1	MIPS-GRPO	4	0.062	64
2	GRPO	1	1.768	1
2	Count IPS-GRPO	4	0.234	64
2	MIPS-GRPO	4	0.061	64

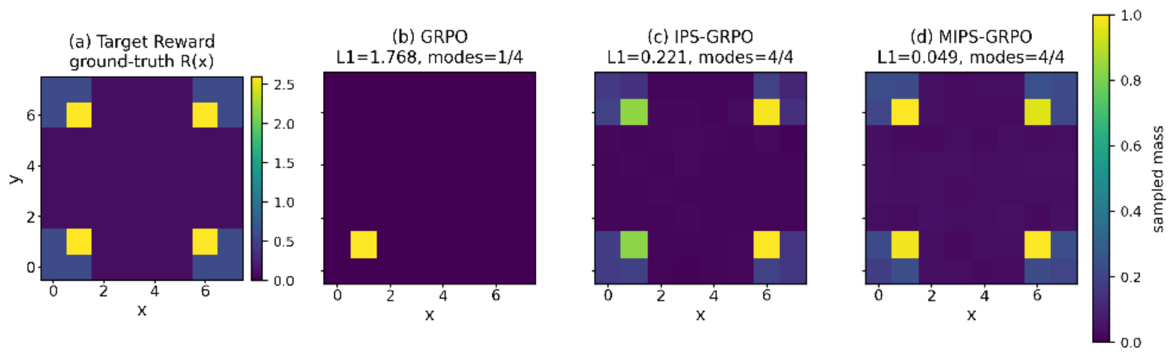


Figure A1: Terminal distributions for the small-grid positive-control regime, $H = 8, G = 256$ (seed 0). This raster was recovered from the previous compiled draft because the original figure link pointed to another machine. Exact values are in the accompanying tables and CSV files.

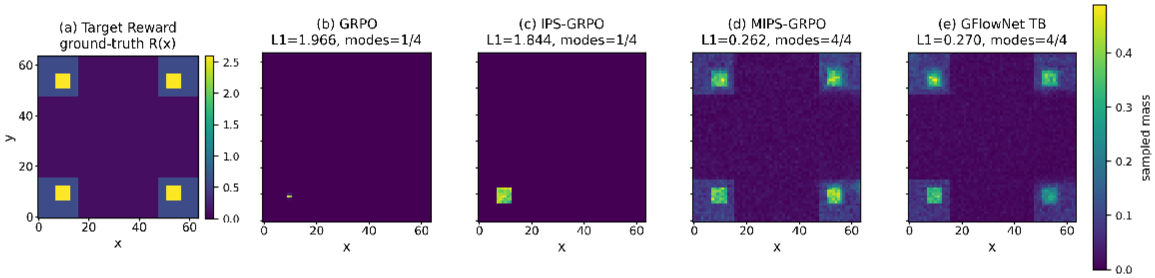


Figure A2: Terminal distributions for the large-grid stress test, $H = 64, G = 32$ (seed 0). GRPO and count IPS-GRPO recover one mode; MIPS-GRPO and GFlowNet TB recover all four.

B Additional phylogenetic diagnostics

The main paper reports the matched four-method five-taxon comparison. Here we retain the older five- and ten-taxon probability–reward views, matched-transform diagnostics, and training curves for context.

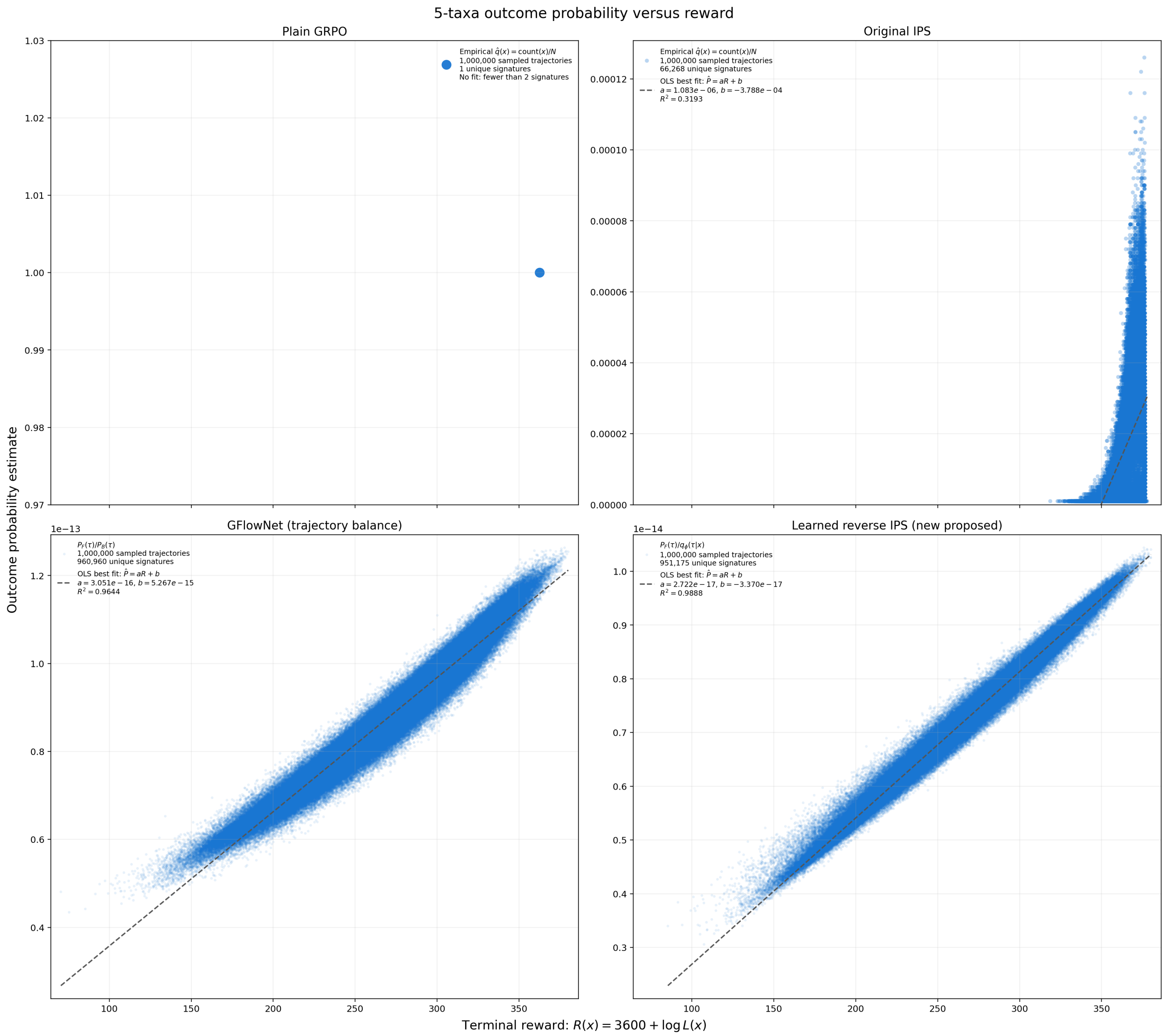


Figure A3: **Five-taxon probability–reward scatter retained for visual context.** This older four-method panel was generated under a mixed legacy protocol (including a replay-based GFlowNet artifact), so it is not the source of the matched no-replay numbers above and should not be used for quantitative ranking.

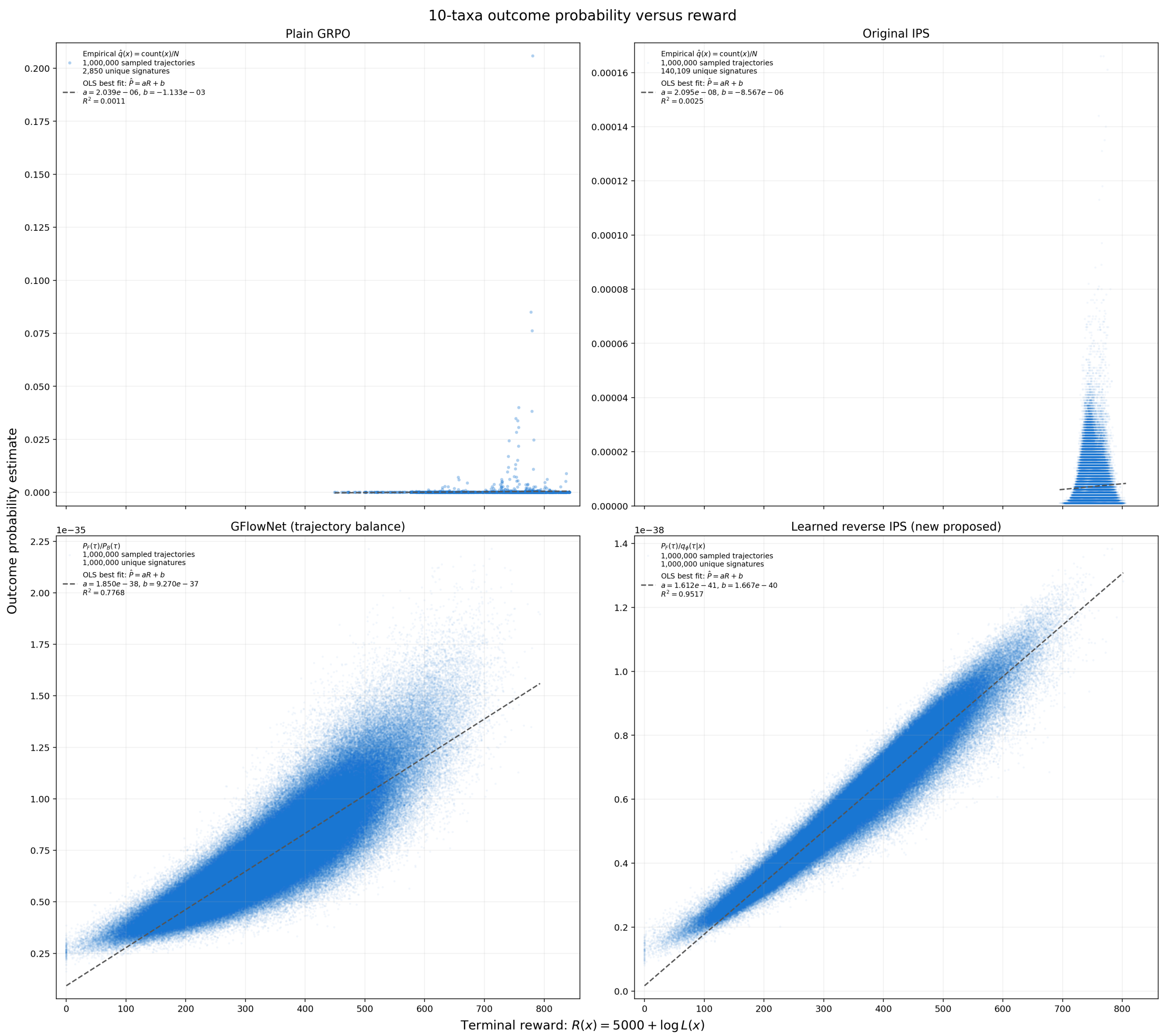


Figure A4: **Ten-taxon scaling diagnostic.** The older four-method probability–reward view shows the collapse signature for GRPO and count IPS and broad sampled support for the multiplicity-aware methods. Because methods sample different terminal sets, this is a within-policy diagnostic rather than a shared-outcome benchmark.

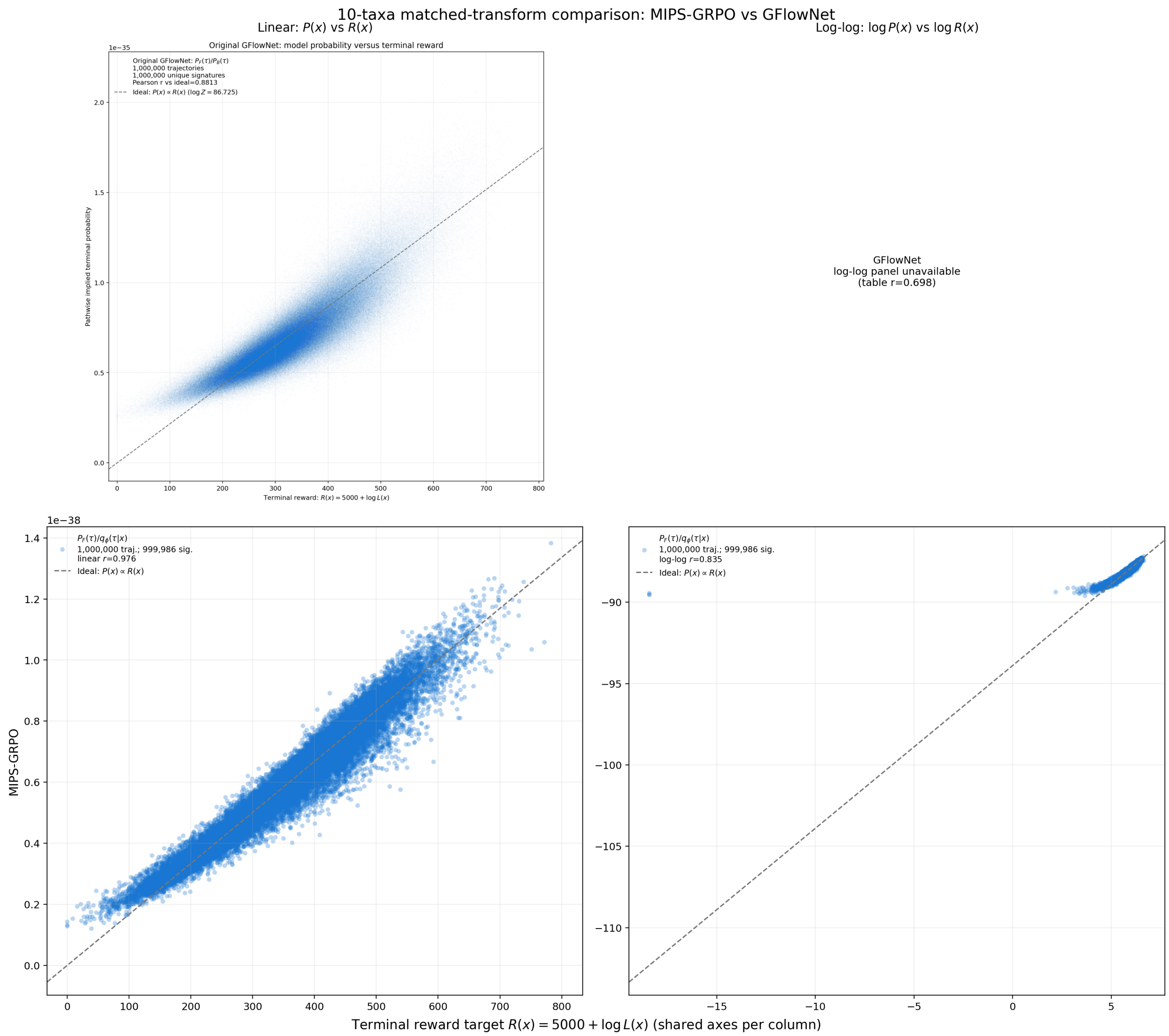


Figure A5: **Ten-taxon matched-transform view.** Linear and log–log transforms are shown consistently across methods. Interpret each correlation as a property of that policy's self-sampled support; a fixed-tree comparison on identical outcomes remains pending.

C Reverse-policy diagnostics

Table A2: Self-sampled pathwise diagnostics on DS1, one million draws per method. These results describe each policy's sampled support and are not fixed-terminal comparisons.

Taxa	Method	Linear r	Log–log r	ESS	Unique / 1M
5	MIPS-GRPO	0.994	0.992	1.000	951,180
5	GFlowNet TB	0.982	0.991	—	100,508
5	Count IPS-GRPO	—	0.565	—	66,268
5	GRPO	—	—	—	1
10	MIPS-GRPO	0.976	0.835	0.995	999,986
10	GFlowNet TB	0.881	0.698	0.977	1,000,000

Table A3: Ten-taxa reverse-policy diagnostic, one million self-sampled trajectories. Fitted and uniform reverse policies have essentially identical ESS, so this experiment does not support a variance-reduction claim.

Method	Reverse policy	Linear r	Log–log r	ESS
MIPS-GRPO	fitted q_ϕ	0.976	0.835	0.995
MIPS-GRPO	uniform	0.000	0.969	0.995
GFlowNet TB	uniform	0.881	0.698	0.977

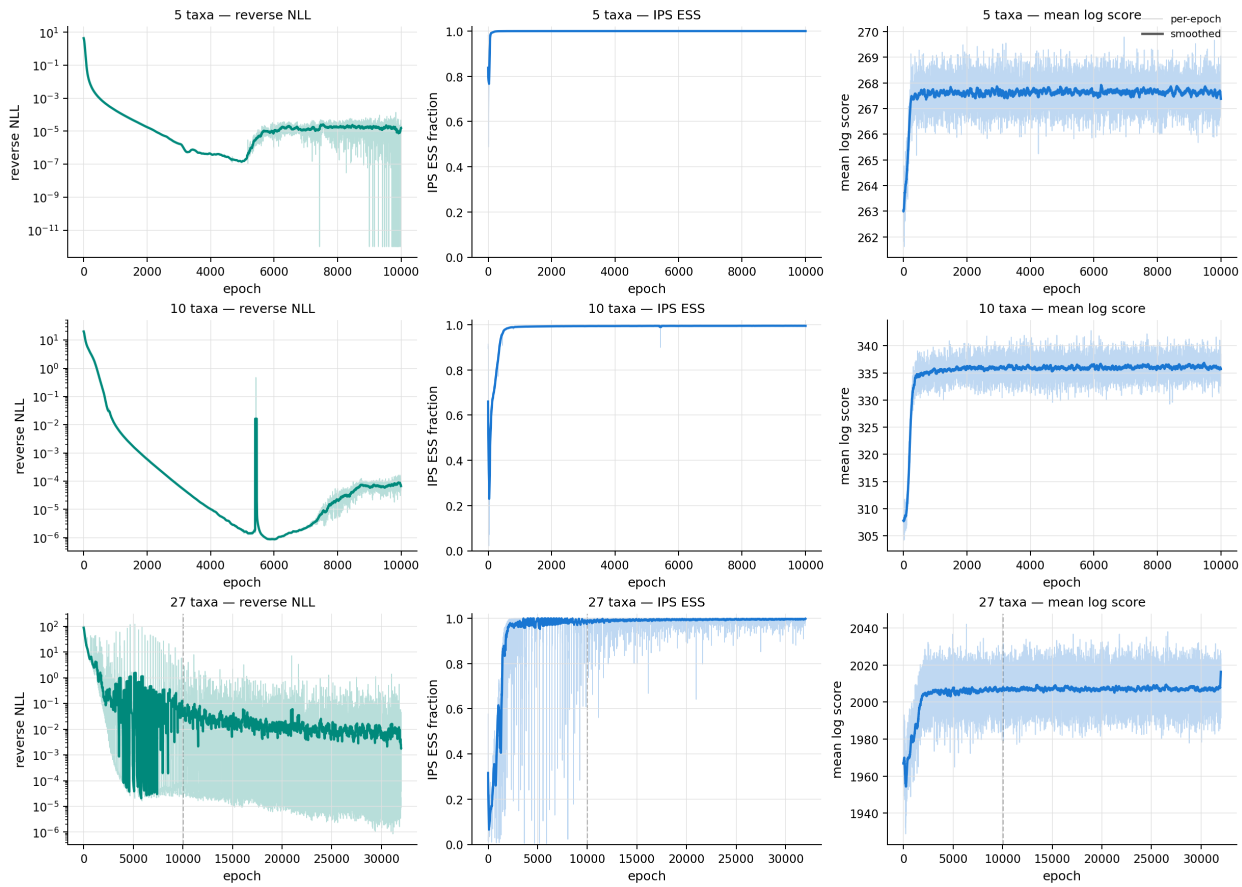


Figure A6: Reverse-policy training curves at 5, 10, and 27 taxa. These curves show training behavior, not final distribution fit.

D Molecular-generation per-seed results

The aggregate molecular result in the main paper is based on the following 12 final-policy evaluations. These values show the substantial seed-to-seed variation in mode discovery that motivates our descriptive interpretation.

Table A4: Per-seed full-space sEH results after 2,500 updates and 50,000 final-policy samples. Unique is the percentage of valid samples that are distinct.

Method	Seed	Valid (%)	Unique (%)	Mean proxy	Modes
GRPO	0	100.00	0.002	3.791	0
GRPO	1	100.00	0.010	4.614	0
GRPO	2	100.00	0.022	4.378	0
Count IPS-GRPO	0	100.00	0.004	5.272	0
Count IPS-GRPO	1	100.00	0.012	3.634	0
Count IPS-GRPO	2	100.00	0.504	2.542	0
MIPS-GRPO	0	99.63	96.272	4.375	42
MIPS-GRPO	1	99.51	91.543	4.392	12
MIPS-GRPO	2	99.79	92.959	4.077	13
RGFN	0	99.83	79.119	4.567	16
RGFN	1	99.99	72.720	3.756	22
RGFN	2	100.00	71.780	4.359	4

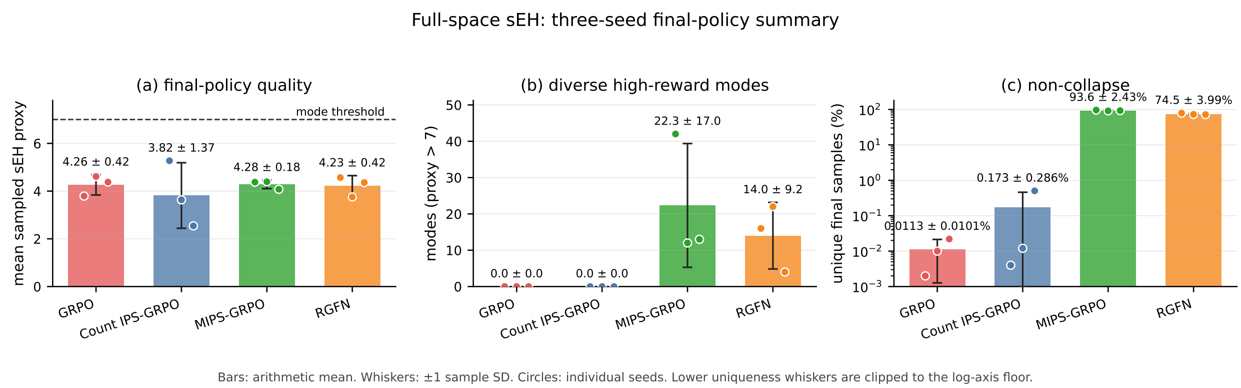
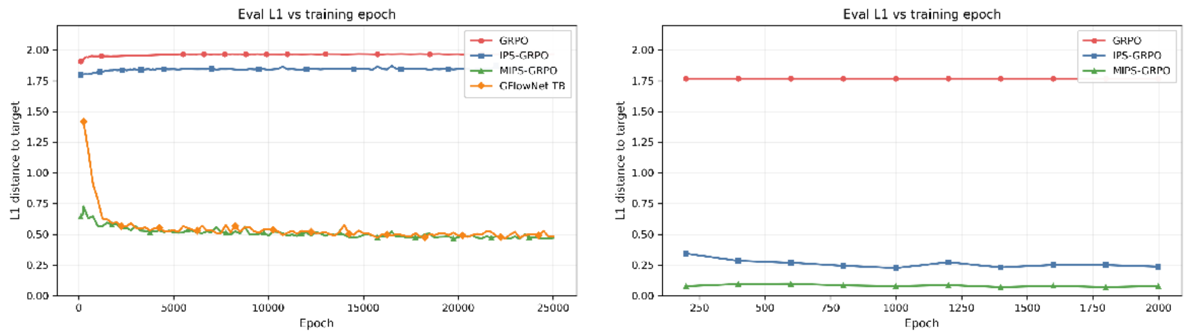


Figure A7: Full-size copy of the three-seed full-space sEH comparison. Bars show arithmetic means, whiskers show ±1 sample standard deviation, and circles show individual seeds.

E Training and additional diagnostics

The following plots document optimization behavior and stricter diagnostics. They are retained as supporting evidence rather than used as primary distributional results.



(a) $H = 64$, $G = 32$ (b) $H = 8$, $G = 256$

Figure A8: Evaluation ℓ_1 during training in both hypergrid regimes. Panels have different scales and should not be compared numerically across grid sizes. The final contrasts are stable rather than isolated checkpoint fluctuations.

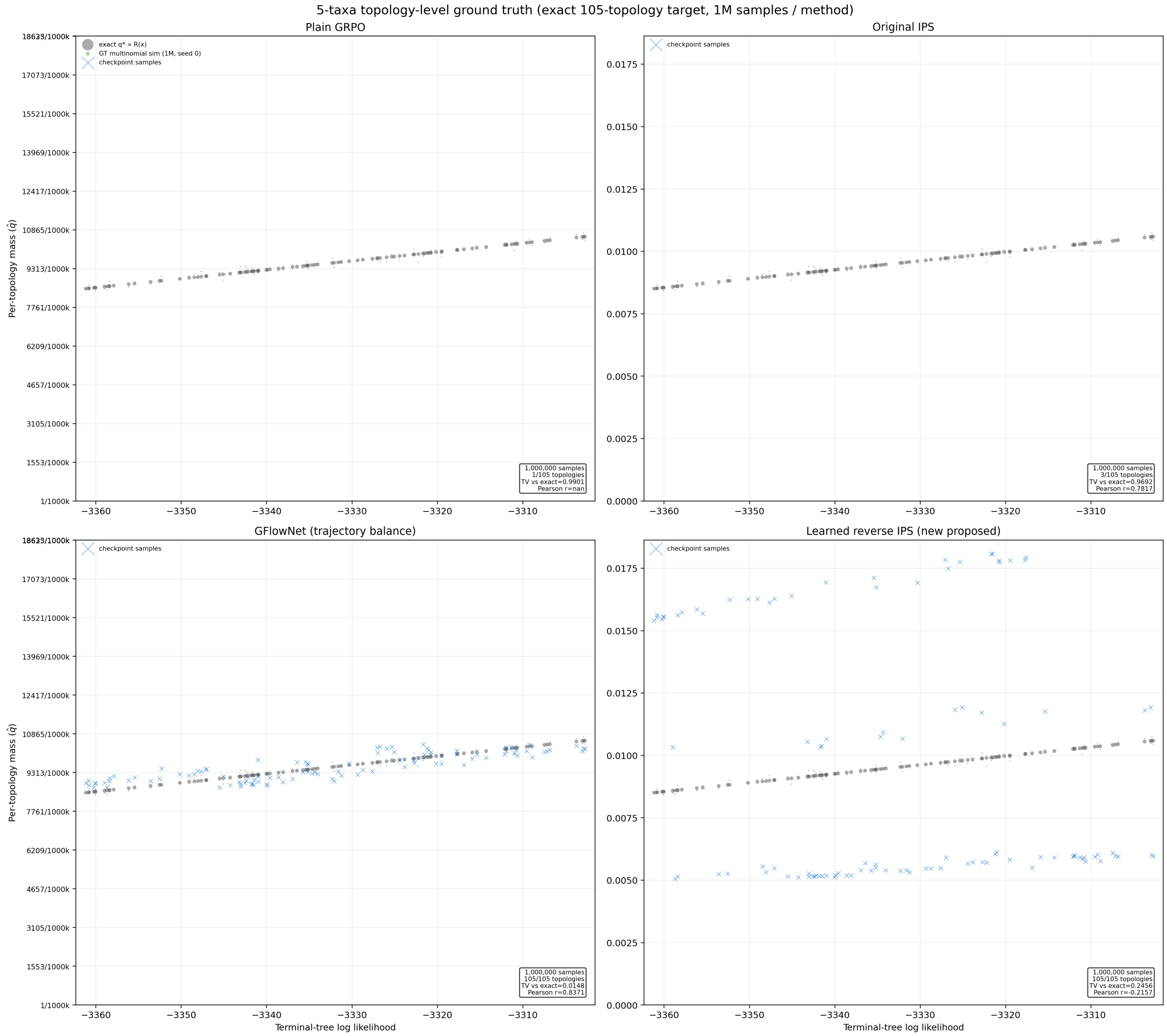


Figure A9: Five-taxa topology-level diagnostic against the exact 105-topology target. This test is stricter than self-sampled signature correlation and exposes topology-level differences hidden by pathwise fits.

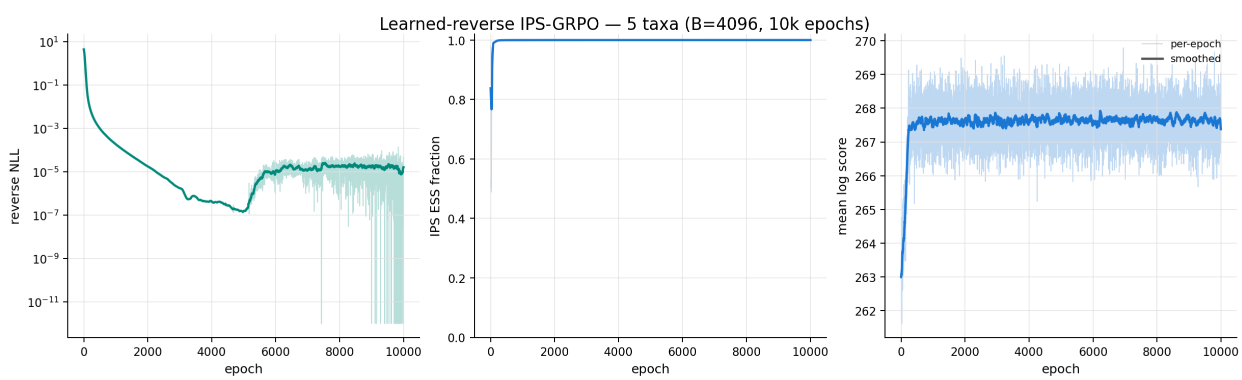


Figure A10: MIPS-GRPO optimization diagnostics at five taxa.

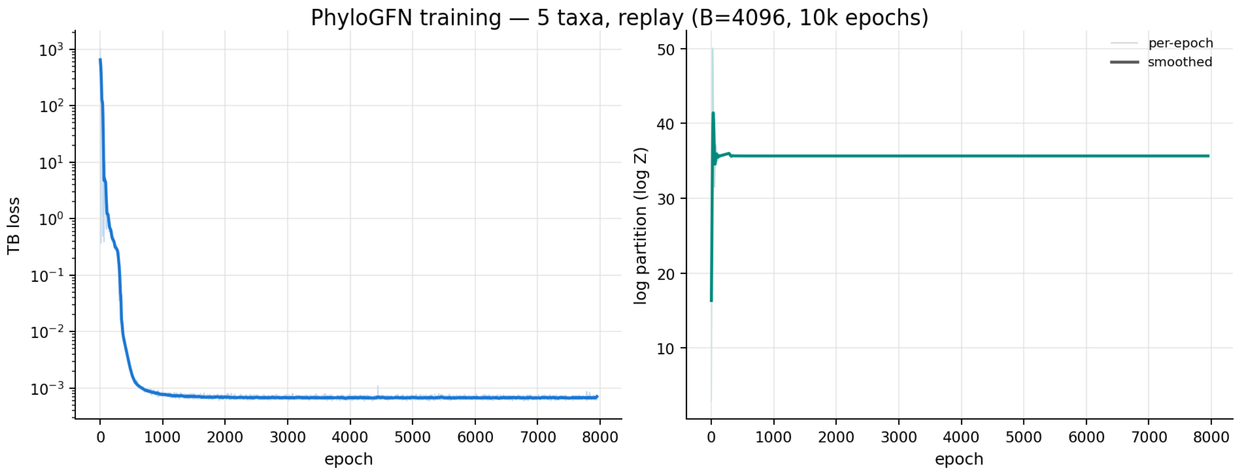


Figure A11: GFlowNet trajectory-balance optimization diagnostics at five taxa.

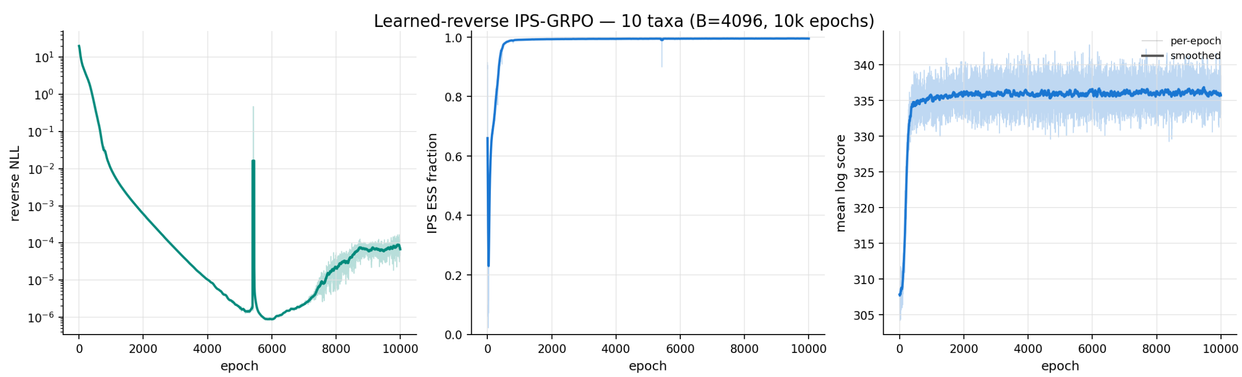


Figure A12: MIPS-GRPO optimization diagnostics at ten taxa.

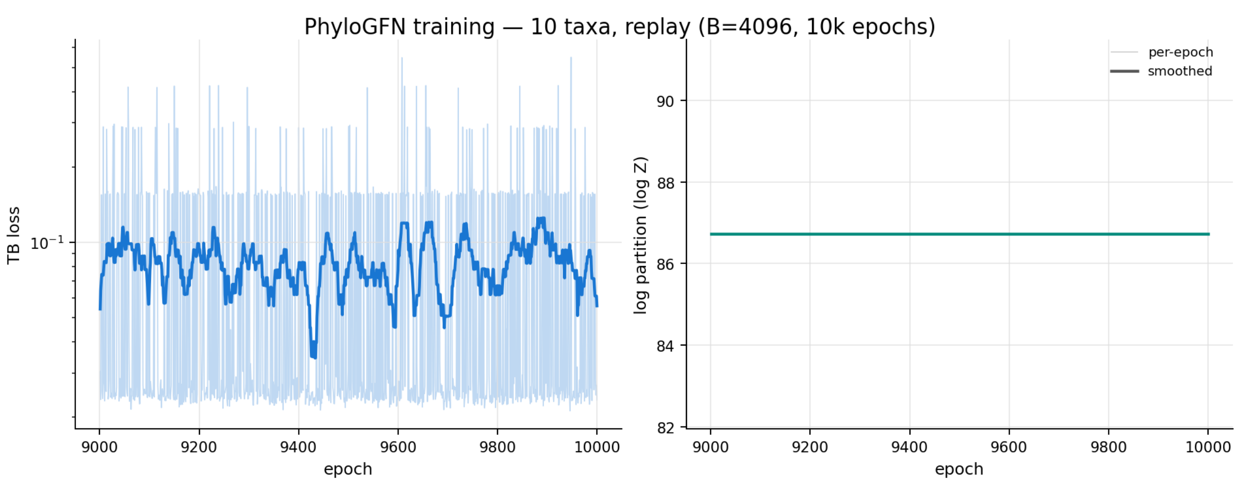


Figure A13: GFlowNet trajectory-balance optimization diagnostics at ten taxa.

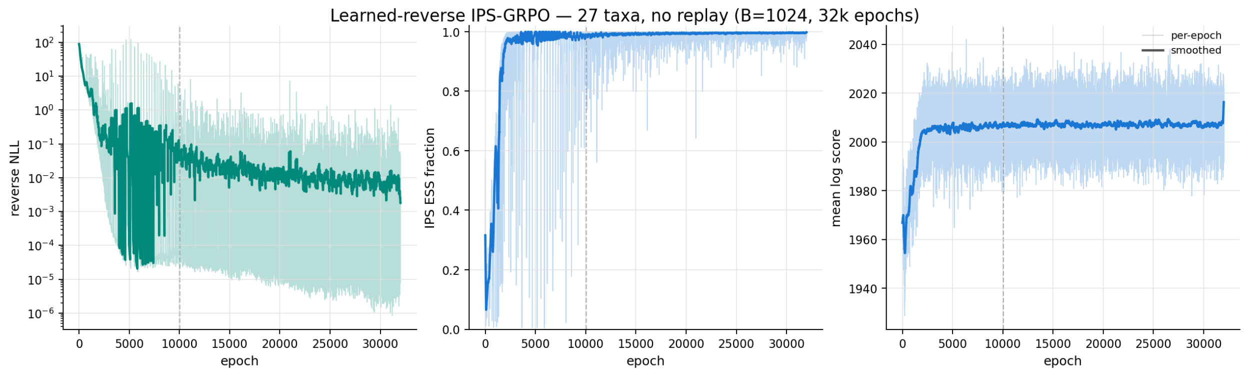


Figure A14: Legacy MIPS-GRPO optimization diagnostics at 27 taxa. The dashed line marks 10,000 epochs.

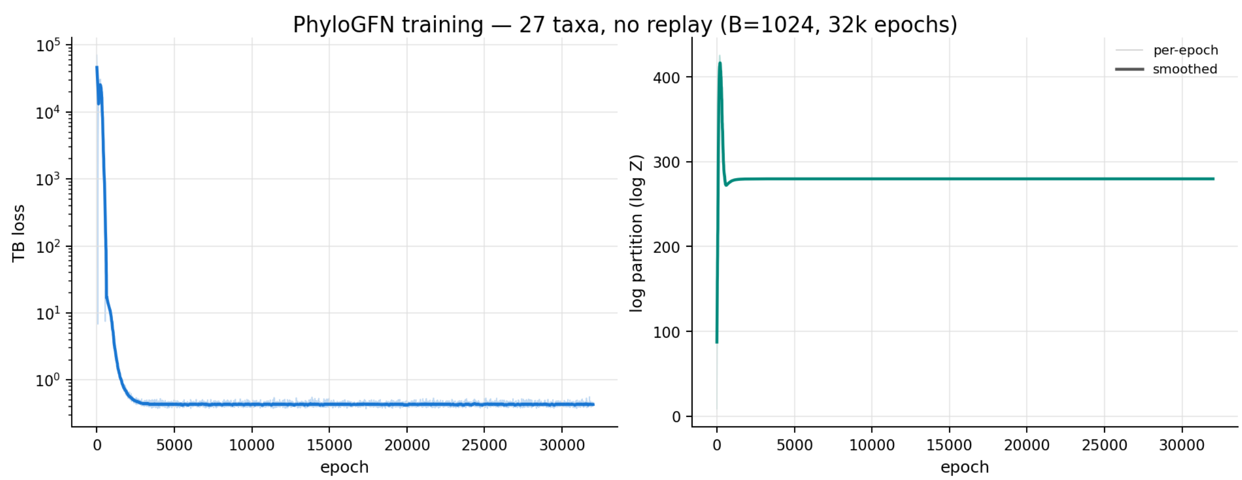


Figure A15: Legacy GFlowNet training curves at 27 taxa. These curves are kept for context and are not part of the main results.

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